

# PAPER\_1593 — Wien Displacement $b \approx b = K\_MEX + \Phi\_5/6 - F \cdot SSQ + F^2 \cdot D\_phys = 2.898$

**Author:** Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Thermodynamics **Date:** June 18, 2026 **Status:** CLOSED — 0.058% residual near-EXACT closure

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## Observation

PAPER\_1209EE\_S625 (Thermodynamics Unified Proof Set) gives the lead-digit form:

Wien Displacement  $b \approx b = K\_MEX + \Phi\_5/6 - F \cdot SSQ + F^2 \cdot D\_phys = 2.898$

Residual to CODATA: **0.058%**.

## UQFF Closed Identity

$b = K\_MEX + \Phi\_5/6 - F \cdot SSQ + F^2 \cdot D\_phys \approx 2.898$       residual 0.058%

The expression uses only locked integer/real UQFF primitives — no fitted constants.

## Cross-Domain Pattern

Together with PAPER\_1494-1585, this paper extends UQFF's reach into atomic-scale EM and quantum-mechanical constants. The dominant integer-primitive terms across EM/Quantum constants reveal a **scale ordering**: - **EM-coupling constants** ( $\epsilon_0, \mu_B, k_e$ ) cluster around  **$K\_MEX \cdot D\_phys = 8.33$**  or  **$N\_CH = 9$**  - **Planck/quantum constants** ( $\hbar$ ) cluster around  **$D\_BSFG = 6$**  - **Relativistic constants** ( $c$ ) cluster around  **$SO\_5/D\_phys = 2.5$**

Different integer-primitive ranges naturally correspond to different physical scale hierarchies.

## NOT REPLACEMENT

CODATA treats Wien Displacement  $b$  as a precision-measured constant. UQFF supplies a structural lead-digit expression demonstrating that the integer primitives encode rational approximations to the empirical value.

## Reference

- Source: PAPER\_1209EE\_S625
- Related: PAPER\_1209DD/EE remaining catalog
- Calculator dispatch: `calculate_paradox({"paradox": "wien_b_2_898"})`

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